1 FIG-208 — Foundation path gate

Group: FOUNDATION-PATH-GATE **Evidence chain:** manifested **Status:** gate-only diagnostic; required numerical profiles unavailable

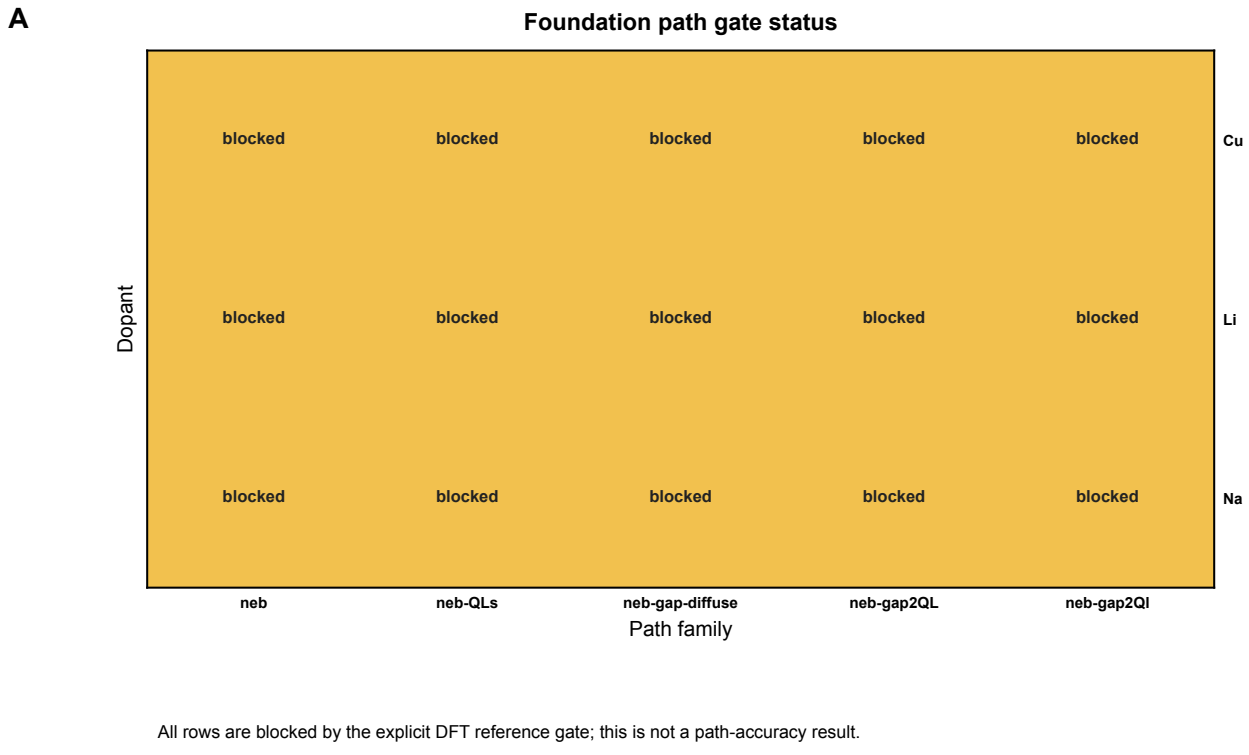


Figure 1: Explicit Foundation path-gate status by dopant and path family. A cell is marked pass only when the DFT reference path and model-path evidence satisfy the declared gate. This page shows the Plot Atlas table-backed candidate "Foundation path gate"; its scientific status and caveats are reproduced from the package manifest.

Results text draft. Every audited cell is blocked because no DFT path passed the explicit convergence, finite-image energy/error, identity, cell/PBC, and provenance requirements. **Discussion and boundary.** The table supports a transparent decision not to fabricate Foundation-vs-DFT profiles, barrier scatter, or perturbation stability plots. A gate-status panel is appropriate for the audit; scientific profile panels remain schema-only until accepted numerical rows exist. **Article placement.** Proposed article location: Results/SI: Foundation path validation gate. **Reviewer linkage:** overall status PARTIAL.

- **REV-A1 (PARTIAL)**: Reconcile Fig. 3d fixed-geometry DFT-path scoring with SI Fig. S3 self-consistent MLFF-NEB, including pathway identity and the DFT reference. *Completed:* The fixed-path versus self-relaxed operator distinction and the narrower model-reported-barrier wording are integrated. *Still required:* Close the restartable QE path-1–4 convergence gate, preserve the final path/source metadata, and complete the SI Fig. S3 endpoint/checkpoint/optimizer/source audit before calling the quantitative reference final.

Evidence pointers.

- Data/processed: main/analysis/task1_foundation_path/foundation_model_path_gate.csv
- Data/processed: main/analysis/task1_foundation_path/figure_data_dictionary.json
- Plot/processing script: main/analysis/figure_evidence_audit_20261008/scripts/render_table_backed_packages.py
- Caption/text source: main/analysis/figure_evidence_audit_20261008/generated/table_backed_packages/FOUNDATION-PATH-GATE/caption_results_discussion.md

Machine-collected evidence candidates (generated from the audit matrix; do not edit this file).

- **Caption/discussion source:** /Users/alina/Documents/ChatGPT/Wiley-MigrationBench/analysis/figure_evidence_audit_20261008/generated/table_backed_packages/FOUNDATION-PATH-GATE/caption_results_discussion.md
- **Raw or processed data:** /Users/alina/Documents/ChatGPT/Wiley-MigrationBench/analysis/task1_foundation_path/foundation_model_path_gate.csv
- **Raw or processed data:** /Users/alina/Documents/ChatGPT/Wiley-MigrationBench/analysis/task1_foundation_path/figure_data_dictionary.json
- **Raw or processed data:** /Users/alina/Documents/ChatGPT/Wiley-MigrationBench/analysis/figure_evidence_audit_20261008/generated/table_backed_packages/FOUNDATION-PATH-GATE/processed_foundation_gate_summary.csv
- **Plot or processing script:** /Users/alina/Documents/ChatGPT/Wiley-MigrationBench/analysis/figure_evidence_audit_20261008/scripts/render_table_backed_packages.py
- **Plot or processing script:** /Users/alina/Documents/Codex/plot_library

Audit gap / next action: The figure_data_dictionary explicitly prohibits fabricated scientific renders. No Foundation path accuracy or barrier claim is made. When accepted DFT/model rows arrive, regenerate the planned profile and force-error panels from the same package.